

WORKING NAME: CUBED

CBM Estimator

Photo-based shipping cost calculator
for African importers

VERSION 1.0 • PLANNING DRAFT • JUNE 2026

Executive Summary

A focused mobile-first tool that lets African importers photograph a supplier's carton and instantly calculate cubic meters (CBM), chargeable weight, total shipping cost, and landed cost per unit.

Today, small importers calculate landed cost with WhatsApp messages, tape measures, and spreadsheets. Mistakes are expensive: a wrong CBM estimate can wipe out margin on an entire shipment or leave the seller quoting customers at a loss.

Our wedge is the capture method: a photo of the carton with a credit card placed on top as a scale reference. Claude vision identifies the carton edges and the card, then computes dimensions. No tape measure, no spreadsheet, no guesswork.

“Snap a photo of your supplier’s carton, get the true landed cost per unit.”

1. Problem Statement

Small importers in West Africa face four compounding problems when quoting landed cost:

Volume estimation is manual and error-prone

Sea freight is priced per cubic meter and air freight by dimensional weight. Importers measure cartons with tape measures, often after the shipment has already arrived. Mistakes compound across multiple carton types.

Suppliers send incomplete information

Carton dimensions are often missing from supplier WhatsApp messages, sent in different units (cm vs. in vs. mm), or buried in a packing list.

Shipping rates are quoted, not centralised

Importers negotiate their own rates with forwarders. There is no single source of truth, and rates change. A tool that hard-codes rates would be wrong for everyone.

Per-unit allocation requires math importers avoid

Total shipping cost is meaningless without per-unit allocation. The math is simple but tedious, especially with multiple carton types in one shipment.

2. Target Customer

Primary

Solo or two-person importers in West Africa (Ghana, Nigeria), 25–40 years old, importing from Guangzhou or Yiwu. They place 1–8 supplier orders a month, use a freight forwarder, and quote landed cost to downstream customers (retail buyers, smaller resellers, or wholesale clients).

Typical profile:

- Revenue between GHS 10,000 and GHS 300,000 per month
- Sources from China via WeChat / Alibaba / WhatsApp
- Ships via established forwarders (Twiga, Bollore, smaller agents)
- Lives in WhatsApp and Excel; mobile-first
- Currently uses a calculator and screenshots for landed-cost math

Secondary (later)

- Sourcing agents (like Priselle) quoting clients
- Freight forwarders quoting their own customers
- Cross-border resellers serving multiple African markets

Not for v1

- Large brands with internal logistics teams
- Established e-commerce on Shopify
- Container-load (FCL) importers — they have forwarders for this

3. Positioning

Core promise

“Snap a photo of your supplier’s carton, get the true landed cost per unit.”

Why customers choose this

- Faster quotes to their customers
- Confidence the math is right
- No tape measure, no spreadsheet
- Works on mobile, in the field, at the port

Competitive landscape

Existing CBM calculators (cbmcalculator.com, Freightos) are basic web forms with no photo capture, no per-unit allocation, no saved history, and no mobile-first UX. Forwarder portals are locked to one service. The gap: a focused mobile tool that captures, calculates, and shares.

4. The Product

v1 does five jobs:

1. **Capture** — get carton dimensions via photo + credit-card reference, OR by typing them in, OR by pasting supplier text
2. **Compute** — total CBM and chargeable weight across all carton types in the shipment
3. **Price** — apply the user's freight rate (sea per CBM, air per kg)
4. **Allocate** — landed shipping cost per unit
5. **Share** — WhatsApp link + PDF export for forwarders, customers, or the importer's records

5. The Capture Mechanism (Differentiator)

The 2-photo method

1. **Top photo.** User places a credit card flat on the top of the carton, photographs straight down. App detects card and carton edges → calculates Length × Width.
2. **Side photo.** User photographs one vertical side of the carton, with the credit card held against or placed beside it. App detects card and carton edge → calculates Height.
3. **Confirm.** App shows estimated dimensions with a confidence score. User taps to accept or edits manually.

Why a credit card

ISO/IEC 7810 ID-1 standard: 85.60 × 53.98 mm. Every importer has one in their wallet. Universal, free, no shipping required.

Backup reference objects:

- A4 paper (210 × 297 mm)
- US Letter (216 × 279 mm)
- Ruler with clearly visible markings
- Hand (lowest accuracy; supported as last resort)

Accuracy expectations

- Two-photo + credit card → **~88–92% accuracy** on standard cartons
- Tolerable for sea-freight quoting (rates round to nearest 0.1 CBM)
- Confidence shown to user: e.g. “Estimated L: 42 cm (±2 cm)”
- Manual override available on every measurement

Quick mode (v1.5)

Single tilted photo with the credit card next to the carton. Vision model infers all three dimensions in one shot. Lower accuracy (~78%) but a single tap. Gated behind a “quick estimate” toggle for experienced users.

Long-term capture options

- **v2:** Paste supplier spec sheet → Claude parses dimensions from text
- **v2:** OCR on supplier packing-list photo
- **v3:** Native iOS LiDAR scan — point and measure, ~95% accuracy

6. End-to-End User Flow

- 1. Open app
- 2. Tap + **New shipment**
- 3. Name the shipment (e.g. “April handbags order”)
- 4. Add carton type #1: take top photo → take side photo → confirm dimensions
- 5. Enter: cartons of this type, units per carton, weight per carton
- 6. (Optional) Add carton type #2 with different dimensions
- 7. Pick freight mode: Sea or Air
- 8. Enter freight rate (e.g. \$480/CBM sea, \$6.50/kg air)
- 9. See results: Total CBM, chargeable weight, total shipping, per-unit cost, sea-vs-air recommendation
- 10. Save / share via WhatsApp link or download PDF

Target: under 90 seconds for a single-carton-type shipment.

7. The Math

All math is deterministic plain code. The LLM only touches the vision step. The calculator never hallucinates.

```
CBM per carton = L(m) × W(m) × H(m)
Total CBM = CBM per carton × number of cartons
Dim weight (kg) = (L cm × W cm × H cm) / 6000 [air freight]
Chargeable (kg) = max(actual_weight, dim_weight)

Sea cost = max(Total CBM × rate per CBM, min charge)
Air cost = max(Chargeable kg × rate per kg, min charge)
Per unit = total cost / (number of cartons × units per carton)
```

8. Screen Map

Route	Purpose
/	Landing page
/app	Dashboard — list of saved shipments
/app/new	New-shipment flow (multi-step, mobile-first)
/app/shipments/[id]	Shipment detail, edit, share
/app/rates	Saved freight-rate presets
/app/account	Billing & settings

Route	Purpose
/s/[shareId]	Public read-only shipment view (no auth)

9. Data Model

Table	Columns
users	id, clerk_id, phone, email, created_at
shipments	id, user_id, name, mode (sea air), freight_rate_id, total_cbm, total_weight, total_cost, per_unit_cost, currency, created_at
cartons	id, shipment_id, length_m, width_m, height_m, weight_kg, count, units_per_carton, photo_top_url, photo_side_url, capture_method, confidence
freight_rates	id, user_id, name, mode, origin, destination, rate, currency, min_charge
ai_jobs	id, user_id, kind, tokens, cost, confidence, raw_response, created_at
shares	id, shipment_id, share_token, view_count, created_at

10. Tech Stack

Layer	Choice
Framework	Next.js 16 App Router on Vercel
Mobile	PWA with HTML5 camera capture (no native app v1)
Styling	Tailwind v4 + shadcn/ui
Auth	Clerk (phone OTP + Google)
Database	Neon Postgres
File storage	Vercel Blob
AI	Vercel AI Gateway → Claude Sonnet 4.6 (multimodal)
Payments	Paystack
PDF generation	@react-pdf/renderer (server-side)
Analytics	Vercel Analytics + PostHog
Errors	Sentry

11. Accuracy & Confidence Strategy

Trust is the make-or-break. Three mechanisms:

- **Confidence score** — every estimate shows expected error in cm, drawn from the vision model's confidence + reference-detection quality
- **Single-tap edit** — every dimension is editable inline; no buried menus
- **Camera guidance** — live overlay shows where to place the card, warns on bad angles or poor lighting
- **“Estimate, not invoice”** labelling on every output; ToS makes liability clear

12. Pricing

Plan	Price	Includes
Free	GHS 0/month	3 shipments/month, manual entry only
Starter	GHS 40/month (~\$3.50)	30 shipments/month, photo capture, save history, WhatsApp share
Pro	GHS 120/month (~\$10)	Unlimited shipments, PDF export, multiple rate presets, history reports
Forwarder (later)	GHS 400/month (~\$33)	Team seats, client-branded shares, multi-user accounts

Photo capture is the paid-tier hook. Free users see the magic in a demo but must subscribe to use it on real shipments.

13. Cost Economics

- Each carton estimate: 2 Claude vision calls
- Sonnet vision @ ~1,500 input + 200 output tokens/call $\approx$ \$0.01 per call $\rightarrow$ **~\$0.02 per carton**
- Pro user: 100 shipments/mo $\times$ 1.5 cartons avg = 150 cartons $\times$ \$0.02 = **\$3.00 AI cost** vs. ~\$10 revenue. Healthy margin.
- Storage and CDN are negligible. Photos downsampled to ~800 $\times$ 800 before upload.

14. Roadmap

v1 (Weeks 1–4)

- Auth, onboarding, freight-rate setup
- 2-photo capture flow with credit-card reference
- Manual entry fallback
- Multi-carton shipments
- CBM & landed-cost calculation
- Save history, WhatsApp share, PDF export
- Billing on Paystack with free tier

v1.5 (Months 2–3)

- Quick mode (single tilted photo)
- Paste supplier spec — Claude parses dimensions
- OCR on supplier packing-list photo
- Per-shipment notes & attachments
- Multi-currency display

v2 (Months 3–6)

- Customs / duty estimator (Ghana first, then Nigeria, Kenya)
- Team accounts (multiple users per workspace)
- Forwarder rate library (community-sourced, opt-in)
- Quote templates & client-branded PDFs

v3 (Months 6+)

- Native iOS app with LiDAR scan
- Sales-rate restock advisor (when paired with catalog data)
- Re-introduce catalog & AI listings as an expansion module

15. What's Not in v1

- Catalog, product pages, storefront
- AI listing, captions, translation
- Customer-facing online checkout
- Duty / customs calculation
- Automatic freight-rate lookups
- LiDAR / AR scanning
- Multi-currency at the shipment level
- Multi-user accounts

16. Risks & Mitigation

Risk	Mitigation
Photo dimensions inaccurate → user under-quotes	Confidence display, easy override, “estimate” labelling
User has no credit card / different reference	Support A4, ruler, coin, hand (degraded accuracy)
Bad lighting, glare, dark cartons	In-app camera overlay with positioning guidance
Irregular cartons (rolls, bags, non-rectangular)	“Bounding box” mode + manual override
Suppliers’ stated weight differs from reality	Show dim weight vs. actual weight side-by-side
Liability if estimate is wrong	Clear “estimate, not invoice” disclaimer; ToS
Free CBM calculators undercut us	Lean on photo + per-unit + save/share — they don’t have these

17. Validation Plan

Two parallel tracks before any production build:

Track A — Customer validation

Interview 5–8 importers in the Priselle network using the script in Appendix A.

- 5+ interviews completed
- 3+ unprompted “I’d pay for this” signals
- 2+ beta-test commitments

Track B — Vision accuracy validation

- Photograph 8–10 real cartons with a credit-card reference (varied lighting, surfaces)
- Measure with a tape for ground truth
- Submit photos to Claude Sonnet 4.6 with structured prompt
- Compare estimates to ground truth, calculate mean error

Decision rule:

- Mean error < 8% → build v1 as planned
- 8–15% → build with mandatory confirmation step
- > 15% → rethink capture method (markers, AR, or pure manual + paste)

18. Open Decisions Before Build

- Final product name (working name: **Cubed**)
- Domain availability and trademark check
- Hosting region (Vercel EU vs. nearest AWS region for African latency)
- Single-founder build vs. design partner
- Beta cohort selection

Appendix A — Importer Interview Script

Use this script to interview 5–8 active importers in the Priselle network. Goal: understand current behaviour and pain, not pitch the product. Follow Mom Test principles — ask about the past, get specifics, avoid hypotheticals.

Logistics

- Format: 15–20 minute call (WhatsApp voice or in-person)
- Record audio with consent (you'll forget the details)
- Take light notes; transcribe key quotes after
- No slides, no demo, no pitch — only listen

Opening (1 min)

“Thanks for the time. I'm doing research on how importers handle the shipping math when they bring goods in from China — what's working, what's a headache. Nothing to sell; I'd just love 15 minutes of your story. Mind if I record audio so I'm not scribbling notes the whole time?”

Question 1 — Current behaviour

Walk me through the last order you placed with a Chinese supplier. From the moment they sent you a quote — what did you do, step by step?

Probes:

- *What did the supplier send you? (Photo, PDF, voice note?)*
- *Where did you write things down? (WhatsApp, Notes, Excel?)*
- *Who else was involved? (Forwarder, partner, customer?)*

Question 2 — The math moment

When the supplier gave you the quote, how did you figure out what it would actually cost you landed in Accra?

Probes:

- *Did you measure the cartons? When — before ordering or after they arrived?*
- *How did you decide between sea and air?*
- *How long did the math take you?*
- *What rate did you use? Where did it come from?*

Question 3 — The worst case

Tell me about a time the landed cost came out way different than you expected. What happened?

Probes:

- *What went wrong? (Wrong measurement, wrong rate, missed duty?)*
- *How much did it cost you?*
- *What did you do differently next time?*

Question 4 — Reaction (only at the very end)

If I told you there was a phone app where you could photograph a carton with a credit card on top, and it told you the CBM, the shipping cost, and the per-unit cost in 30 seconds — what's the first thing you'd want to know?

Probes:

- *Would you trust the photo measurement? Why / why not?*
- *What would have to be true for you to actually use it?*
- *Who else on your team would touch it?*

Question 5 — Willingness to pay

If this worked the way you're imagining, how much would you pay for it per month?

Probes:

- *(After they answer) What would push you to pay more / less than that?*
- *Compare to anything else you pay for monthly in your business.*

Closing

"Really helpful. Last question — would you be open to being one of the first people to try it when I have something to show? No commitment, just an early look."

Recording template

After each interview, fill in:

- Date, name, role, location
- Orders per month, average shipment value, ship method
- Current workflow (in 3 sentences)
- Top 3 pain points named
- Best quote captured verbatim
- Willingness to pay: stated number
- Beta interest: yes / maybe / no

Decision framework after 5 interviews

Signal	Action
3+ unprompted "I'd pay for this"	Build v1 as planned
1–2 unprompted "I'd pay"	Build a stripped-down prototype first; re-validate with same cohort
0 unprompted "I'd pay"	Pause. Reframe problem. Talk to a different customer (forwarders?)
Strong signal for a different feature (duty calc, supplier translation)	Note for v1.5 but stay focused on CBM core

Common pitfalls to avoid

- Pitching instead of listening — people are polite, not honest
- Asking hypothetical questions (“Would you use...”) instead of past behaviour
- Accepting compliments as validation — only money or commitments are real
- Talking to friends and family — they want you to succeed, so they lie nicely
- Skipping the bad-case question — the worst story is where the real pain lives